Class-Imbalance Mitigation on BRACS with a LoRA-Adapted Foundation-Model Encoder

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Abstract

A frozen Virchow2 encoder with a logistic-regression readout loses 7.65 points of balanced accuracy on BRACS when the training classes are imbalanced at a ratio of 100. This report asks how much of that loss six established mitigation methods recover when the encoder adapts through low-rank updates (LoRA). On 30 patient-disjoint split-draw shards, a cross-entropy (CE) model is trained on a balanced class allocation, and CE and every method are trained on allocations with ratios 50 and 100 and with BRACS's native class shares; each method's strength is selected per shard on validation patients. Under LoRA, CE loses 4.95 points at ratio 100 and 3.54 points at ratio 50, while its 0.85-point loss at native shares cannot be distinguished from zero. At ratio 100, a curriculum of data augmentation (CUDA) gives the most accurate model, 2.59 points above CE, or about half of the loss. Train-time and post-hoc logit adjustment follow at about two-fifths; together with classifier-only GCL, whose second-stage sampler is close to class-balanced, they are the only methods that substantially raise recall of the rarest classes. No method closes the gap. Fitting the methods on the balanced allocation as well shows how they behave without imbalance: CUDA improves the balanced model by 1.39 points, so most of its gain does not depend on imbalance, whereas GCL and cRT lower balanced accuracy slightly and are less affected by imbalance than their gain over CE suggests. Except for CUDA, methods that re-fit only the classifier on a frozen encoder recover as much as methods trained end to end. The smaller loss under LoRA reflects a balanced model about three points less accurate than the frozen readout, not a more robust imbalanced one.

1 Motivation

When the patient support per class is held fixed, class imbalance alone reduces balanced accuracy on BRACS by 7.65 points at an imbalance ratio of $\rho = 100$, against 2.91 points on TCGA-UT, for a logistic-regression readout on frozen Virchow2 features (Queisler, 2026c,d). On BRACS, this loss splits into two effects of similar size (Queisler, 2026a). In the *frequency* effect, the common classes dominate the training loss and pull the decision boundary toward themselves. In the *support* effect, the rare classes offer too few distinct patches to describe how they vary, which no reweighting can supply. These analyses keep the encoder frozen, so they do not test the established long-tailed methods that also shape the representation.

Six such methods have been defined for patch classification under a Virchow2 encoder adapted only through LoRA (Queisler, 2026b): balanced sampling and a curriculum of data augmentation (CUDA) at the data level, logit adjustment and DisAlign at the algorithm level, and classifier re-training (cRT) and Gaussian clouded logit (GCL) adjustment as hybrids. This report asks: *once the encoder can adapt, how much of BRACS's imbalance loss does each method recover, and does recovery differ between methods trained end to end and methods that freeze the encoder and re-fit only the classifier?*

2 Design

The shards follow the imbalance design of Queisler (2026c). BRACS has three fixed patient-disjoint splits, each with about 103 training, 24 validation, and 24 test patients, and each split is drawn ten times. A draw samples $G = 10$ training patients per lesion class from those with at least 160 training patches and holds a total budget of $T = 7 \cdot G \cdot 32 = 2{,}240$ patches fixed across all allocations. Every allocation of a draw uses the same patients and the same random assignment of classes to allocation ranks, so the 30 split-draw shards differ between allocations only in the class shares. Training reads decoded patch images, because every method adapts the encoder: a rank-8 LoRA update on the attention projections plus a linear head (a cosine head for GCL), as defined by Queisler (2026b).

2.1 Allocations and methods

Each shard's patch budget is divided among the seven classes in four ways, following an exponential allocation profile. The *balanced* allocation ($\rho = 1$) trains CE, which serves as the *undamaged reference*, and the methods whose effect there is not zero by construction (Section 3). Three imbalanced allocations, with ratios $\rho = 50$ and $\rho = 100$ and with BRACS's native class shares (N, realized ratio about 6.7), each train CE as their *damaged reference* and every mitigation method over the grid of Table 1. The two ratios show whether recovery grows with severity; the native shares ask whether mitigation matters at the imbalance BRACS actually has.

Ratio 100 is close to the most severe imbalance this design can express. At $\rho = 100$ the rarest class receives about 12 of the 2,240 patches, roughly one per patient. Beyond $\rho \approx 125$ it would receive fewer patches than its ten patients, so the tail class could no longer draw on all its patients and the design would mix imbalance with patient shortage.

Methods fall into two training families. *End-to-end* methods (balanced sampling, CUDA, train-time logit adjustment, GCL) train the LoRA update and head together. CUDA (Ahn et al., 2023) augments image patches before the encoder, with a per-class strength that rises as the current model learns the class, so it can only act end to end. *Classifier-only* methods freeze an end-to-end model's encoder and re-fit only classifier parameters: post-hoc logit adjustment, DisAlign, and cRT start from the allocation's CE model, and GCL's classifier-only form, its second training stage, starts from the end-to-end GCL model of the same σ. Logit adjustment and GCL therefore each appear once per family, giving eight method variants.

The strength grids bracket the defaults of the original methods. Balanced sampling and CUDA's augmentation probability end at their maximum of 1, and train-time τ at the full prior correction $\tau = 1$. The other grids extend past any edge on which the validation selection would otherwise concentrate, which at native shares widens the GCL and DisAlign grids. GCL's scale is $s = 30$, and its second-stage sampler uses the effective-number range $a = 0.9999$, $b = 10^{-5}$ of the reference implementation.

2.2 Training length and fit count

End-to-end training length was fixed in advance on CE alone. CE was trained on the balanced and the $\rho = 100$ allocation, one draw per split, for 150, 300, 600, and 1,200 steps, and the rule selects the smallest step count within 0.5 points of the best mean validation balanced accuracy at $\rho = 100$. The four candidates reached 31.87, 34.22, 34.93, and 33.42 percent, so 600 steps (batch 256, learning rate 10^{-4}, cosine schedule) were used for all end-to-end fits; classifier-only fits train 200 steps. One length serves all allocations, so that a recovery difference between allocations cannot come from training length. It was chosen at the most severe allocation and not re-tuned at $\rho = 50$ or native shares. Per shard, the grid holds 39 end-to-end and 41 classifier-only configurations, for 1,170 and 1,230 fits over the 30 shards. Every configured fit completed and entered the analysis.

Training	Method	Strength	Grid
End to end	CE (reference)	–	–
End to end	Balanced sampling	s	0.5, 1.0
End to end	CUDA	p_{aug}	0.25, 0.5, 1.0
End to end	Logit adjustment (train-time)	τ	0.5, 1.0
End to end	GCL	σ	0.02, 0.05, 0.2 (N: also 0.5)
Classifier only	Logit adjustment (post hoc)	τ	0.5, 1.0, 1.5, 2.0
Classifier only	DisAlign	γ	0.5, 1.0, 1.5, 2.0 (N: also 3.0)
Classifier only	cRT	–	–
Classifier only	GCL	σ of its source	as GCL

Table 1: Strength grids of the imbalanced allocations. CUDA's acceptance threshold (0.6), check size (10), and maximum strength (10) are fixed, and its class levels are updated every nine steps. The balanced allocation trains CE, CUDA over its grid, both GCL forms at $\sigma = 0.05$, DisAlign at $\gamma = 1$, and cRT.

3 Evaluation

The primary endpoint is test patient-macro balanced accuracy (BA), the mean of per-class patch recalls in which every test patient of a class carries equal weight. Estimates pool the 30 shards with equal weight per split. Uncertainty combines Bayesian bootstrap weights on test patients (Rubin, 1981) with resampling of draws within each split over 2,000 replicates. All allocations and methods share the replicate weights, so every contrast is paired; intervals are 95 percent percentile intervals.

For each method, shard, and imbalanced allocation, the grid value with the best *validation* BA is selected; the test partition is never read for selection. With m^* the selected fit and a the allocation,

$$D_a = \mathrm{BA}(\mathrm{CE}, 1) - \mathrm{BA}(\mathrm{CE}, a), \qquad R_m = \mathrm{BA}(m^*, a) - \mathrm{BA}(\mathrm{CE}, a), \qquad \mathrm{share}_m = R_m / D_a, \quad (1)$$

where the share is computed per replicate. A recovery counts as *material* if R_m is at least one point and its interval lies above zero, and as *full* if the interval of $\mathrm{BA}(m^*, a)$ contains the point estimate of $\mathrm{BA}(\mathrm{CE}, 1)$.

R_m compares a method with CE, so it mixes two effects: a correction of the imbalance and any general effect the method has on accuracy. To separate the two, the methods are also fitted on the balanced allocation with the same validation selection. A method's *balanced gain* and its *imbalance-specific recovery* are

$$G_m = \mathrm{BA}(m^*, 1) - \mathrm{BA}(\mathrm{CE}, 1), \qquad R_m - G_m = D_a - \big[\mathrm{BA}(m^*, 1) - \mathrm{BA}(m^*, a)\big], \quad (2)$$

the difference between the damage CE suffers and the damage the method suffers. At equal class counts, balanced sampling and both forms of logit adjustment are identical to CE, so $G_m = 0$ for them by construction and they need no balanced fits. GCL's cloud size and DisAlign's reweighting are also uniform there, so their strengths have no effect and one value each is fitted; GCL then reduces to CE with a cosine head.

Secondary readings are descriptive: head, body, and tail recall by assigned class rank; macro negative log-likelihood (NLL) and expected calibration error (ECE), raw and after validation temperature scaling; the frequency of each selected strength; and the mean share of the end-to-end and classifier-only families. The damage of the frozen readout (Queisler, 2026c) uses the same test patients but a different classifier, so it serves as an external reference without a paired interval.

	$\rho = 50$ ($D = 3.54$)		$\rho = 100$ ($D = 4.95$)		N ($D = 0.85$)
Method	R_m	share	R_m	share	R_m
CUDA	**1.81** [0.65, 2.97]	0.51 [0.20, 0.91]	**2.59** [1.42, 3.79]	0.52 [0.29, 0.82]	**1.21** [0.34, 2.14]
Logit adj. (train-time)	**1.12** [0.08, 2.13]	0.32 [0.04, 0.58]	**1.96** [0.79, 3.17]	0.40 [0.19, 0.60]	−0.17 [−0.99, 0.63]
Logit adj. (post hoc)†	**1.45** [0.40, 2.52]	0.41 [0.15, 0.71]	**1.94** [0.79, 3.13]	0.39 [0.18, 0.59]	−0.07 [−0.49, 0.36]
GCL	**1.27** [0.26, 2.23]	0.36 [0.09, 0.66]	**1.48** [0.58, 2.41]	0.30 [0.12, 0.51]	−0.01 [−1.03, 1.02]
GCL†	**1.37** [0.07, 2.67]	0.39 [0.03, 0.73]	**1.95** [0.46, 3.50]	0.39 [0.12, 0.64]	−0.55 [−1.48, 0.33]
Balanced sampling	0.72 [−0.11, 1.57]	0.20 [−0.05, 0.43]	**1.14** [0.32, 1.98]	0.23 [0.07, 0.41]	−0.02 [−0.89, 0.80]
cRT†	0.65 [0.11, 1.20]	0.18 [0.04, 0.33]	0.88 [0.11, 1.62]	0.18 [0.03, 0.30]	−0.16 [−0.67, 0.37]
DisAlign†	0.61 [0.26, 1.05]	0.17 [0.08, 0.32]	0.58 [0.28, 0.93]	0.12 [0.06, 0.18]	0.15 [−0.09, 0.39]

Table 2: Six variants recover at least one point at $\rho = 100$; no share interval reaches the full damage. Recovery R_m in BA points and share of the CE damage, with paired 95% intervals; bold marks material recovery ($R_m \geq 1$, interval above zero). †Classifier only, frozen encoder; unmarked rows train end to end. Shares at native shares are omitted because the damage there cannot be distinguished from zero.

4 Results

4.1 Damage under LoRA

The balanced CE model reaches 36.53 percent BA [33.82, 39.45]. It loses $D = 3.54$ points [1.93, 5.21] at $\rho = 50$ and 4.95 points [3.25, 6.74] at $\rho = 100$. At native shares the damage is 0.85 points [−0.31, 2.08], an interval that includes zero. All three damages are smaller than under the frozen readout, which lost 6.63, 7.65, and 4.05 points on the same test patients (Queisler, 2026c). The reason is the undamaged reference, not the imbalanced allocations: LoRA's balanced model is about three points less accurate than the frozen readout (36.53 against 39.57 percent), while the imbalanced models land close to the frozen values (32.99 against 32.94 at $\rho = 50$, 31.57 against 31.92 at $\rho = 100$, 35.67 against 35.51 at N).

4.2 Recovery per method

Figure 1 and Table 2 give each method's paired recovery. At $\rho = 100$, six of eight variants recover a material part of the damage: CUDA (2.59 points [1.42, 3.79]), train-time and post-hoc logit adjustment (1.96 and 1.94), classifier-only and end-to-end GCL (1.95 and 1.48), and balanced sampling (1.14). cRT (0.88) and DisAlign (0.58) recover a positive amount below the one-point threshold. CUDA's share of the damage, 0.52 [0.29, 0.82], is the largest, followed by logit adjustment's 0.40 [0.19, 0.60]; no share interval reaches 1, so part of the damage remains under every method. CUDA alone formally reaches full recovery at $\rho = 100$: the interval of its accuracy, [31.70, 36.81], just contains the balanced 36.53 percent, while its paired share places the remaining gap between about one-fifth and seven-tenths of the damage.

At $\rho = 50$ the ordering is the same with smaller gains. CUDA (1.81 points, share 0.51 [0.20, 0.91]), logit adjustment (both forms), and GCL (both forms) recover a material part, the latter four with shares of 0.31 to 0.41, and balanced sampling falls just short (0.72, interval includes zero). These five variants also formally reach full recovery, because the interval of an unpaired accuracy is about five points wide, wider than the 3.54-point damage; their paired shares, whose intervals end below 0.92, show that the gap is not closed. At native shares, seven of eight recoveries lie between −0.55 and 0.15 points with an interval that includes zero. CUDA is the exception: it gains 1.21 points [0.34, 2.14] and reaches 36.89 percent, above the balanced CE model's 36.53, although the damage it would recover, 0.85 points, cannot be distinguished from zero.

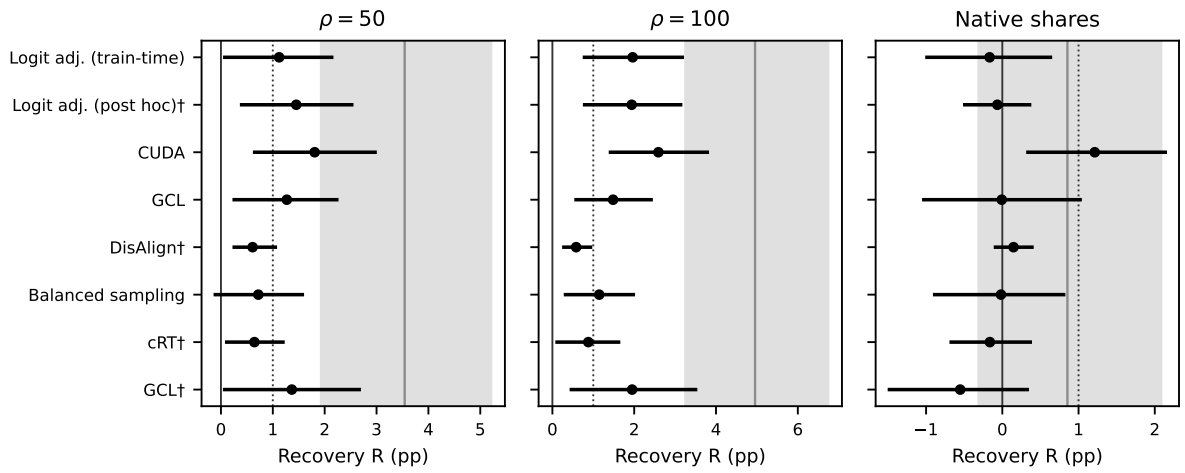


Figure 1: CUDA, logit adjustment, and GCL recover one to three points at $\rho = 50$ and $\rho = 100$, short of the damage; at native shares only CUDA moves. Paired recovery R_m with 95% intervals. The grey line and band mark the CE damage D with its interval, the solid line zero, and the dotted line the one-point threshold.

		$R_m - G_m$			$\rho = 100$
Method	G_m	$\rho = 50$	$\rho = 100$	N	share
CUDA	1.39 [0.40, 2.37]	0.41 [−1.07, 1.95]	1.20 [−0.16, 2.62]	−0.18 [−1.39, 1.13]	0.24 [−0.04, 0.49]
GCL	−0.33 [−1.25, 0.53]	**1.60** [0.22, 2.95]	**1.81** [0.72, 3.00]	0.32 [−1.15, 1.94]	0.37 [0.17, 0.59]
GCL†	−0.64 [−1.62, 0.20]	**2.01** [0.48, 3.63]	**2.59** [0.93, 4.37]	0.09 [−1.32, 1.54]	0.52 [0.23, 0.77]
cRT†	−0.48 [−0.90, −0.11]	**1.12** [0.45, 1.81]	**1.36** [0.56, 2.19]	0.31 [−0.35, 1.03]	0.27 [0.13, 0.41]
DisAlign†	−0.01 [−0.06, 0.03]	0.62 [0.26, 1.06]	0.59 [0.29, 0.94]	0.16 [−0.08, 0.40]	0.12 [0.06, 0.18]

Table 3: Net of their effect on the balanced allocation, GCL and cRT correct more of the imbalance than their recovery against CE shows, and CUDA less. Balanced gain G_m and imbalance-specific recovery $R_m - G_m$ (Eq. 2) in BA points, with its share of the CE damage at $\rho = 100$; paired 95% intervals, bold marks material recovery. †Classifier only. Balanced sampling and logit adjustment have $G_m = 0$ by construction, so their R_m in Table 2 is already imbalance-specific.

4.3 Imbalance-specific recovery

The balanced gains point in opposite directions (Table 3). CUDA reaches 37.92 percent BA on the balanced allocation, a gain of $G = 1.39$ points $[0.40, 2.37]$ over CE. GCL (-0.33) and its classifier-only form (-0.64) lose accuracy there, both with intervals that include zero, and cRT loses 0.48 points $[-0.90, -0.11]$; DisAlign is indistinguishable from CE (-0.01).

Net of these gains, CUDA's imbalance-specific recovery at $\rho = 100$ is 1.20 points $[-0.16, 2.62]$, a share of 0.24 of the damage, and is not material; at $\rho = 50$ it is 0.41 and at native shares -0.18 points. The other methods correct more than their recovery against CE shows. Classifier-only GCL corrects 2.59 points $[0.93, 4.37]$ at $\rho = 100$, a share of 0.52 $[0.23, 0.77]$, the largest of any method, and end-to-end GCL 1.81 points. cRT's correction, 1.36 points $[0.56, 2.19]$, is material, and DisAlign's stays at 0.59. At $\rho = 100$ six of eight variants are material on this reading as well: both forms of logit adjustment and balanced sampling, whose G_m is zero, both forms of GCL, and cRT. CUDA, which led against CE, drops out. On the balanced allocation, validation selects CUDA's $p_{\text{aug}} = 0.5$ in 17 and 0.25 in 13 of 30 shards, and never 1.

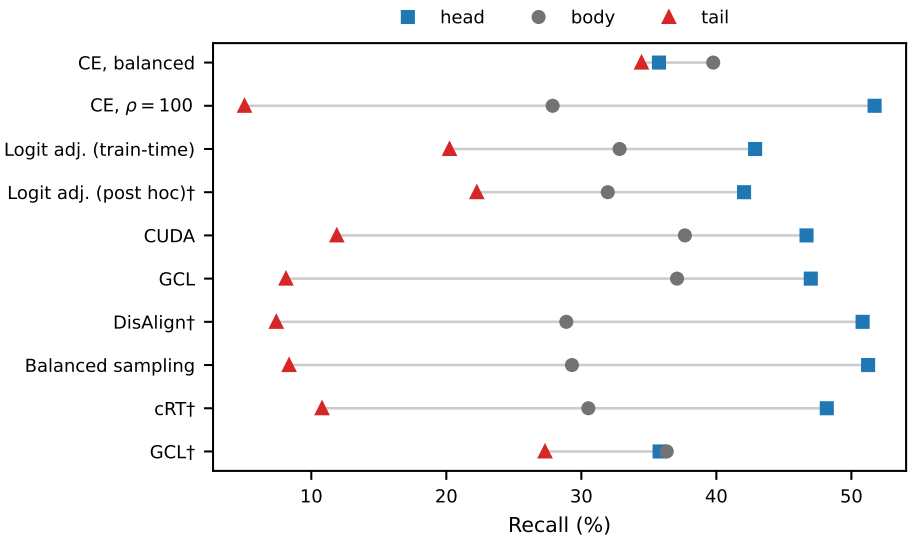


Figure 2: Logit adjustment and classifier-only GCL lift tail recall substantially at $\rho = 100$; CUDA and end-to-end GCL gain in the body, the other methods stay close to CE. Patient-macro recall of the head, body, and tail thirds of classes by assigned rank, averaged over shards.

4.4 End-to-end versus classifier-only training

The two training families differ mainly through CUDA, which has no classifier-only form. The mean share is 0.36 for end-to-end and 0.27 for classifier-only methods at $\rho = 100$, and 0.35 and 0.29 at $\rho = 50$; without CUDA, the end-to-end mean at $\rho = 100$ is 0.31. The matched pairs agree most closely: post-hoc logit adjustment, applied to the CE model at inference without retraining, recovers as much as its train-time form (1.94 against 1.96 points at $\rho = 100$), and re-fitting GCL's classifier adds 0.47 points to end-to-end GCL at $\rho = 100$ and 0.10 at $\rho = 50$. These family and pair comparisons are point estimates without intervals.

4.5 Where recovery lands

At $\rho = 100$ the CE model keeps 5.1 percent recall in the tail third of classes by assigned rank, against 34.5 percent when balanced, and gains 16 points in the head (Figure 2). Logit adjustment and classifier-only GCL move the tail substantially, to 20.2 (train-time) and 22.3 percent (post hoc) for logit adjustment, paid for by about nine points of head recall, and to 27.3 percent for classifier-only GCL, whose sampler is close to class-balanced: its head recall, 35.8 percent, equals the balanced model's 35.7, and its body recall, 36.3 percent, is 3.5 points below it. End-to-end GCL and CUDA recover through the body instead, 37.1 and 37.7 against 27.9 percent. CUDA raises the tail to 11.9 percent and gives up five points of head recall; end-to-end GCL, balanced sampling, cRT, and DisAlign leave the tail at 7 to 11 percent. Rank-group recalls are point estimates.

4.6 Selected strengths

Validation selection mostly picks interior or theoretically motivated values. At $\rho = 100$, logit adjustment selects the full correction $\tau = 1$ in 23 of 30 shards in its train-time form, and $\tau = 1$ is also the post-hoc mode (15 of 30, with 0.5 in 11). Balanced sampling selects full balancing $s = 1$ in 22 of 30. GCL selects $\sigma = 0.05$ in 15 (end to end) and 13 (classifier only, tied with 0.02) of 30, and DisAlign selects $\gamma = 1$ or 1.5 in 22 of 30. CUDA's selected augmentation probability rises with the imbalance: $p_{\text{aug}} = 1$ wins 14 of 30 shards at $\rho = 100$, 13 of 30 at $\rho = 50$, and none

at native shares, where 0.25 wins 17. The other selections at $\rho = 50$ resemble those at $\rho = 100$. At native shares the selections spread across the grid; DisAlign's five values each win between 4 and 8 shards. With no resolved damage, validation accuracy is nearly flat in the strength, and the spread reflects noise rather than a preferred value.

4.7 Probability quality

Imbalance mainly worsens raw calibration, and temperature scaling removes most of it for every method. At $\rho = 100$ the CE model's raw NLL rises from 2.89 to 3.49 nats and its raw ECE from 28.4 to 36.4 points; even the balanced LoRA model is strongly overconfident. CUDA has by far the lowest raw NLL, 2.25 nats $[2.10, 2.41]$, and raw ECE, 20.8 points $[16.7, 26.4]$, below even the balanced CE model (2.89 and 28.4); the two logit-adjustment forms follow (2.99 and 2.94 nats), and cRT (4.64) and balanced sampling (3.98) are highest. After scaling, every method lies between 1.71 (CUDA) and 1.82 nats, around CE's 1.79 and close to the balanced 1.67, and scaled ECE falls from 12.3 for CE to between 9.3 and 12.3 points. The scaled intervals are unpaired and overlap: CUDA's augmentation curbs raw overconfidence, but no method changes probability quality beyond what temperature scaling already does.

5 Discussion

How much does each method recover? At $\rho = 100$ CUDA gives the most accurate model, recovering about half of the damage against CE, followed by logit adjustment at about two-fifths; six of eight method variants recover a material part. None closes the gap: no paired share interval reaches 1. The ranking is stable from $\rho = 50$ to $\rho = 100$, and the absolute gains grow with the damage.

What do the balanced fits add? For the quality of the trained classifier, recovery against CE is the relevant reading: it is what a practitioner gains by switching methods on imbalanced data. The balanced fits add how each method behaves without imbalance, and the imbalance-specific recovery follows from that as how much less a method is damaged than CE. The two readings differ only for methods that change accuracy on balanced data, and the difference runs both ways. GCL and cRT make the balanced model slightly worse, cRT by a resolved half point, and are less damaged by the imbalance than their recovery suggests; on this reading classifier-only GCL corrects about half of the damage. The subtraction assumes that the balanced effect carries over unchanged to the imbalanced allocations.

Is CUDA's gain an imbalance correction? Mostly not, but it is still the best classifier. CUDA improves the LoRA model by 1.39 points on the balanced allocation, and this general gain explains its 1.21-point gain at native shares and more than half of its 2.59-point recovery at $\rho = 100$. The remaining 1.20 points are not resolved from zero. The point estimate and one descriptive observation leave room for a smaller imbalance-specific part: the selected augmentation probability rises from 0.25 or 0.5 on the balanced allocation to 1 in about half the shards at $\rho = 50$ and $\rho = 100$. As a correction of imbalance, CUDA's measured contribution is smaller than that of logit adjustment, GCL, and cRT; as a way to train a better LoRA model on BRACS, it is the most effective method tested.

Does it matter whether the encoder adapts? Not for correcting the imbalance. Post-hoc logit adjustment, a fixed prior subtraction at inference, matches every end-to-end method except CUDA, whose recovery exceeds it by 0.65 points in point estimate but whose imbalance-specific part, 1.20 points, falls below it. The largest imbalance-specific correction, 2.59 points, comes from classifier-only GCL, which re-fits only the classifier on a frozen encoder. Without CUDA, the classifier-only family recovers as much as the end-to-end one. Its share under LoRA, about 0.4, exceeds the quarter it recovered under the frozen readout (Queisler, 2026a), but the two numbers rest on different damages and classifiers. The methods that act directly on class

frequency, logit adjustment and GCL's near-balanced sampler, are also the only ones that restore tail recall, which fits the large frequency effect found on BRACS under the frozen readout.

Does LoRA itself help? No. LoRA shrinks the measured damage from 7.65 to 4.95 points at $\rho = 100$ only because its balanced model is about three points worse than the frozen readout; its imbalanced models are as accurate as the frozen ones. The best mitigated model at $\rho = 100$, CUDA at 34.17 percent, stays 5.4 points below the frozen balanced readout. At BRACS's native class shares, LoRA's damage cannot be distinguished from zero and no rebalancing method differs from CE, so in this pipeline rebalancing matters only at severe imbalance; CUDA's gain there matches its gain on the balanced allocation and is a general improvement of the LoRA model.

Limitations. The study covers one dataset, one encoder, and one adaptation scheme. End-to-end training length was tuned on CE at $\rho = 100$ only, and a method or allocation that converges more slowly, such as CUDA under strong augmentation, may be disadvantaged at 600 steps. Imbalance-specific recovery assumes that a method's effect on the balanced allocation carries over additively to the imbalanced ones, which the design does not test. The full-recovery reading compares an unpaired interval with a point estimate and is weak whenever the damage is smaller than that interval, as at $\rho = 50$; the paired share is the informative reading. The comparison with the frozen readout is descriptive, because the two pipelines differ in classifier and training and share only the test patients.

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